

Aligned CIFAR-10 Semantic Codebook

Shared transmitter-receiver initialization for representation drift experiments

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85.36%	100%	0.0
CODEBOOK TEST ACCURACY	TX-RX AGREEMENT	INITIAL SEMANTIC DRIFT

Technical summary

Phase 1 now has an auditable aligned semantic codebook. The epoch-9 baseline checkpoint was used to encode only the 45,000-image training split into 128-D representations. One L2-normalized mean prototype was computed for each CIFAR-10 class, then cloned into independent transmitter and receiver artifacts. The aligned prototype path achieved **85.36% test accuracy**, retained **98.04%** of the linear classifier accuracy, and produced **zero initial drift** with 100% endpoint agreement.

The completed aligned reference

Element	Phase 1 aligned-codebook definition
Source model	Best CIFAR-10 checkpoint: epoch 9, seed 42
Representation	128-D output of the trained projection head
Codebook	10 L2-normalized class prototypes; one per CIFAR-10 class
Construction data	45,000 training images only; deterministic evaluation transforms
Decoder rule	Cosine nearest prototype
Endpoint state	Independent files with exactly equal initial values; no local updates

PHASE 1 STATUS: COMPLETE, ALIGNED, AND REPRODUCIBLE

One training-only codebook initializes both endpoints

The transmitter first maps an image to a 128-D semantic vector. The nearest normalized class prototype is selected by maximum cosine similarity. That prototype is the semantic message, and the receiver interprets it against an equal but independently stored copy of the codebook. No endpoint parameter or prototype is updated in this phase.



$$q^* = \operatorname{argmax}_k \cos(z, c_k)$$

At initialization: $C_{T,0} = C_{R,0}$

$$c_k = \operatorname{normalize}(\operatorname{mean}\{z_i : y_i = k\}) \quad | \quad q^* = \operatorname{argmax}_k \cos(z, c_k)$$

Construction controls protect the test set

Control	Implementation	Reason
Training-only prototypes	45,000 train images	Validation and test samples cannot influence codebook values.
Deterministic transforms	Resize, center crop, ImageNet normalization	No random crop or flip changes the extracted reference vectors.
Frozen checkpoint	Epoch 9; model.eval()	Codebook generation does not train or update model weights.
Normalized comparison	L2 prototypes plus cosine similarity	Vector direction, rather than scale, determines the nearest class.

All ten classes are represented

Class	Train n	Class	Train n
Airplane	4,470	Dog	4,503
Automobile	4,517	Frog	4,483
Bird	4,522	Horse	4,482
Cat	4,522	Ship	4,489
Deer	4,507	Truck	4,505

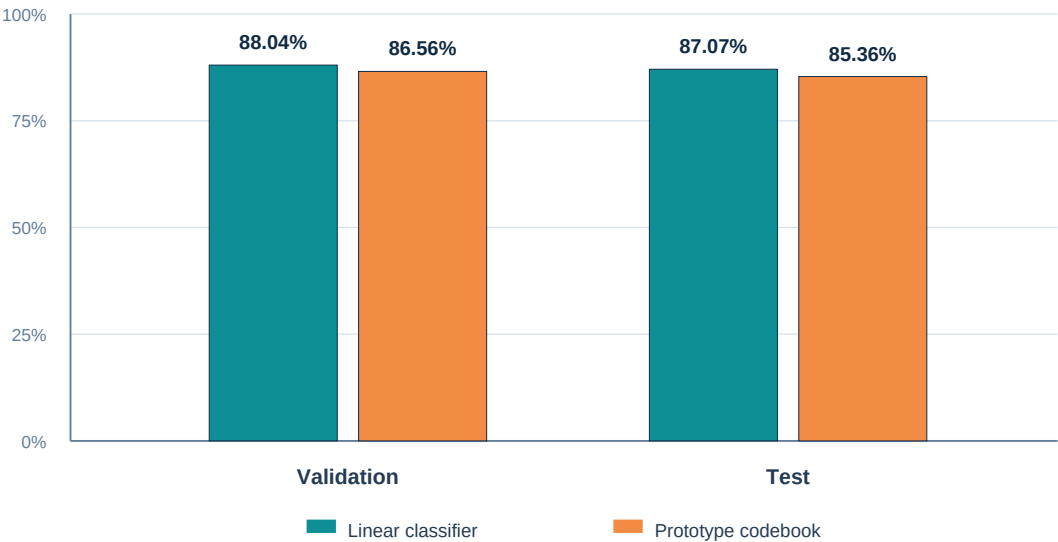
Counts range from 4,470 to 4,522 and sum to 45,000; every class prototype is supported by thousands of training observations.

Prototype decoding preserves 98.04% of classifier accuracy

The prototype decoder is intentionally simpler than the trained linear classifier. On the 10,000-image test set it scored **85.36%**, a **1.71 percentage-point** reduction from 87.07%. This small, measured cost creates an interpretable 10-item semantic codebook and defines the aligned reference L0 for later drift experiments.

Classifier and codebook accuracy

Correct predictions; validation n=5,000 and test n=10,000



Split	Classifier	Codebook	Difference	Retention	n
Validation	88.04%	86.56%	-1.48 pp	98.32%	5,000
Test	87.07%	85.36%	-1.71 pp	98.04%	10,000

How to interpret the comparison

The original classifier is retained as the model-quality check: its re-evaluated 88.04% validation and 87.07% test accuracies exactly match the baseline report. The codebook path is evaluated separately because it replaces the learned 128-to-10 decision layer with nearest-prototype decoding. Future degradation must therefore be measured from **85.36%**, not from the 87.07% classifier result.

Aligned codebook baseline: L0 = 85.36% test accuracy

The prototype space is structured rather than collapsed

The normalized prototypes are distinct: off-diagonal cosine similarities range from 0.319 to 0.872, with a mean of 0.543. Related visual categories occupy nearby directions; cat-dog is the closest pair. This is descriptive evidence of class structure, not proof that geometry alone explains every classification error.

Prototype cosine-similarity matrix

Normalized 128-D class prototypes; diagonal similarity equals 1.00



Cosine similarity



Airplane	1.00	0.59	0.58	0.47	0.50	0.39	0.39	0.53	0.81	0.62
Automobile	0.59	1.00	0.36	0.42	0.33	0.41	0.36	0.41	0.61	0.80
Bird	0.58	0.36	1.00	0.75	0.79	0.68	0.78	0.66	0.44	0.35
Cat	0.47	0.42	0.75	1.00	0.72	0.87	0.74	0.67	0.41	0.42
Deer	0.50	0.33	0.79	0.72	1.00	0.68	0.68	0.83	0.38	0.33
Dog	0.39	0.41	0.68	0.87	0.68	1.00	0.62	0.69	0.34	0.40
Frog	0.39	0.36	0.78	0.74	0.68	0.62	1.00	0.50	0.37	0.32
Horse	0.53	0.41	0.66	0.67	0.83	0.69	0.50	1.00	0.39	0.42
Ship	0.81	0.61	0.44	0.41	0.38	0.34	0.37	0.39	1.00	0.63
Truck	0.62	0.80	0.35	0.42	0.33	0.40	0.32	0.42	0.63	1.00
	air	auto	bird	cat	deer	dog	frog	horse	ship	truck

Pair	Cosine similarity	Interpretation
Cat - Dog	0.872	Highest off-diagonal similarity
Deer - Horse	0.835	Second-highest similarity
Airplane - Ship	0.805	Third-highest similarity

Endpoint alignment is exact by construction

The canonical tensor was cloned twice, not independently estimated. The transmitter and receiver files therefore have the same numerical checksum while remaining separate artifacts that can be updated independently in a later phase. All initial drift measures are zero and every transmitted prototype decodes to the same receiver index.

Integrity check	Observed value	Pass condition
Codebook shape	10 x 128	Expected dimensions
Exact tensor equality	True	True
Independent storage	True	True
Maximum absolute difference	0.0	0.0
Mean cosine distance	0.0	0.0
Mean Euclidean distance	0.0	0.0
Tx-Rx agreement	100.00%	100.00%
Tensor SHA-256	144bf74665c951e4...7574ab3c5599	Same at both endpoints

What this phase establishes - and what it does not

The result establishes a clean common semantic state before any asymmetric learning, channel corruption, detector, or realignment operator is introduced. It does not yet show that drift will occur, that a receiver-side signal can detect it, or that a low-overhead update can recover accuracy. Those are the questions for the next experimental phases.

Limitation	Why it matters	Required follow-up
Single random seed	The accuracy is a reproducible run, not a variance estimate.	Run 3-5 seeds and report mean plus standard deviation.
One prototype per class	A single mean compresses within-class modes and may merge related classes.	Compare multiple prototypes or a VQ codebook after the minimal pipeline is stable.
No channel or drift yet	This report validates initialization, not robustness or detection.	Add the abstract channel and asymmetric local updates next.
Frozen backbone	The result is a controlled budget baseline, not maximum CIFAR-10 accuracy.	Keep it fixed for the first drift study; fine-tune only as an ablation.

Recommended next steps

- Freeze the aligned artifacts.** Treat the saved epoch-9 checkpoint and both codebook files as the immutable t=0 reference.
- Add the abstract task-message channel.** Support quantization, feature dropout, erasure, and additive embedding noise as controlled options.
- Inject asymmetric local drift.** Update transmitter and receiver states from different local streams while keeping the channel error-free first.
- Measure degradation from L0.** Log accuracy, cosine prototype drift, Euclidean prototype drift, and task-disagreement rate per round and seed.